Nathaniel Felsky-Deelstra

Kingston, Ontario | Nathanielfd5@gmail.com | 226-583-6001

linkedin.com/in/nathaniel-felsky-deelstra

Summary

First-year Computer Engineering student with a passion for embedded systems, electronics, and team-based engineering design. Strong foundation in C/C++ and hardware prototyping, with practical experience in sensor integration, interference mitigation, and leadership. Seeking a summer engineering position to apply problem-solving, initiative, and technical skills.

Education

Queen's University, BASc Computer Engineering

Clubs and Teams

Electrical Member, Aquatonomous — September 2025 – Current

- Developed interference-mitigation strategies for the telemetry subsystem using magnetic field calculations and EM shielding principles.
- Designed sensor, radio, and wiring layout to reduce noise and improve signal reliability.
- Conducted research on Faraday cages and magnetic field attenuation to inform design decisions.

Elected Year Representative, Queen's ECE Executive Club — September 2025 – Current

- Acted as liaison for the 1st year CE and the ECE Executive, gathering feedback and communicating issues.
- Designed a first-year program patch and supported branding.
- Assisted in planning and running community-building events, including ECE Night and intramural activities.

Elected Treasurer, First Year Engineering Executives — September 2025 – Current

- Managed an annual budget and ensured compliance with Engineering Society financial policy.
- Tracked spending, updated budgets, and ensured a surplus was maintained
- Supported planning for merchandise sales and student events.

Experience

Team member, Tim Hortons — December 2023 – June 2025

- Worked in a team to efficiently complete orders and keep wait time down, while ensuring customer satisfaction.

Projects

Motion Activated Camera

- Built a wireless motion-activated camera with rechargeable power, cloud alerts, and onboard processing.
- Integrated PIR sensors, an ESP32/Arduino controller, and custom power regulation circuitry.
- Applied soldering, voltage regulation, and micro-controller programming to assemble a functional prototype.

Water Pre-Treatment Plant

- Designed a turbidity-reduction system achieving more than 2500 NTU's improvement using coagulation and press filtration in under 8 minutes.
- Developed mechanical parts using CAD and calculated gear ratios for accurate meshing and motor speed
- Implemented Arduino-based automation and wiring for pump and sensor control.

Skills

Languages: C++, C, C#, Python

Software: Autodesk Fusion, Jupyter Notebook, Excel, Word, Visual Studio Code, Arduino IDE